

Assignment No	14
Title	Pentaho Operation
Objective	1. Calculator Operation 2. Concatenation Operation 3. Split Operation 4. Number range Operation 5. String Operation 6. Importing csv file 7. Merge Join 8. Data Validation
Roll No	MCA2511

1. Calculator Operation

Input

Rows of step: Table input (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Transformation

Calculator

Step name

Calculator

☒ Throw an error on non existing files

Fields:

#	New field	Calculation	Field A	Field B	Field C	Value type	L
1	total	A + B	EMPNO	SALARY		None	

Output

Rows of step: Table output (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO	total
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101	40001.0
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101	30002.0
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103	20003.0
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102	15004.0
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103	35005.0
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101	15007.0
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102	28008.0
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101	32009.0
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103	36010.0
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104	41011.0
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101	29012.0
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102	22013.0
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104	37014.0
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103	43015.0

2. Concatenation Operation

Input

Examine preview data

Rows of step: Table input 2 (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Transformation

Concat fields

Step name: Concat fields

Target Field Name: Concatenated

Length of Target Field: 0

Separator: ;

Enclosure: "

Insert TAB

Fields Advanced

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Trim Type	Null
1	EMPNO	Integer	0						both	
2	NAME	String							both	
3	POSITION	String							both	

Get Fields Minimal width

OK Cancel

Help



Output

Rows of step: Table output 2 (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO	Concatenated
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101	1;abc ;xyz
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101	2;e2 ;p1
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103	3;e3 ;p2
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102	4;e4 ;p1
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103	5;e5 ;p3
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101	7;e6 ;p3
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102	8;e7 ;p2
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101	9;e8 ;p1
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103	10;e9 ;p3
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104	11;e10 ;xyz
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101	12;e11 ;p1
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102	13;e12 ;p2
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104	14;e13 ;p3
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103	15;e14 ;xyz

3. Split Operation

Input

Examine preview data

Rows of step: Table input (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Close Show Log

Transform

Split fields

Step name: Split fields

Field to split: NAME

Delimiter:

Enclosure:

Fields

#	New field	ID	Remove ID?	Type	Length	Precision	Format	Group	Decimal	Currency	Nullif	Default	Trim type
1	First Name		N	String	20								none
2	Last Name		N	String	20								none

Help OK Cancel

Output

Examine preview data

Rows of step: Table output (14 rows)

#	EMPNO	First Name	Last Name	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	<null>	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	<null>	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	<null>	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	<null>	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	<null>	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	<null>	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	<null>	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	<null>	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	<null>	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	<null>	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	<null>	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	<null>	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	<null>	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	<null>	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Close

4. Number Range Operation

Input

Rows of step: Table input (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Transformation

Number range

Step name: Number range

Input field: EMPNO

Output field: range

Default value(if no range matches): unknown

Ranges (min <= x < max):

#	Lower Bound	Upper Bound	Value
1	5.0	10.0	5-10
2		5.0	Less than 5
3	10.0		More than 10

OK Cancel

Help



Output

Rows of step: Table output (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO	range
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101	Less than 5
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101	Less than 5
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103	Less than 5
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102	Less than 5
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103	5-10
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101	5-10
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102	5-10
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101	5-10
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103	More than 10
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104	More than 10
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101	More than 10
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102	More than 10
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104	More than 10
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103	More than 10

5. String Operation

Input

Examine preview data

Rows of step: Table input (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Close Show Log

Transformation

String operations

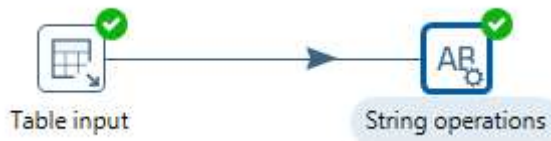
Step name String operations

The fields to process:

#	In stream field	Out stream field	Trim type	Lower/Upper	Padding	Pad char	Pad Length	InitCap	Escape	Digits	Remove Special character
1	NAME		none	lower	left	*	5	N	None	none	none
2	POSITION		none	lower	right	-	5	N	None	none	none

Help OK Get fields Cancel

Output



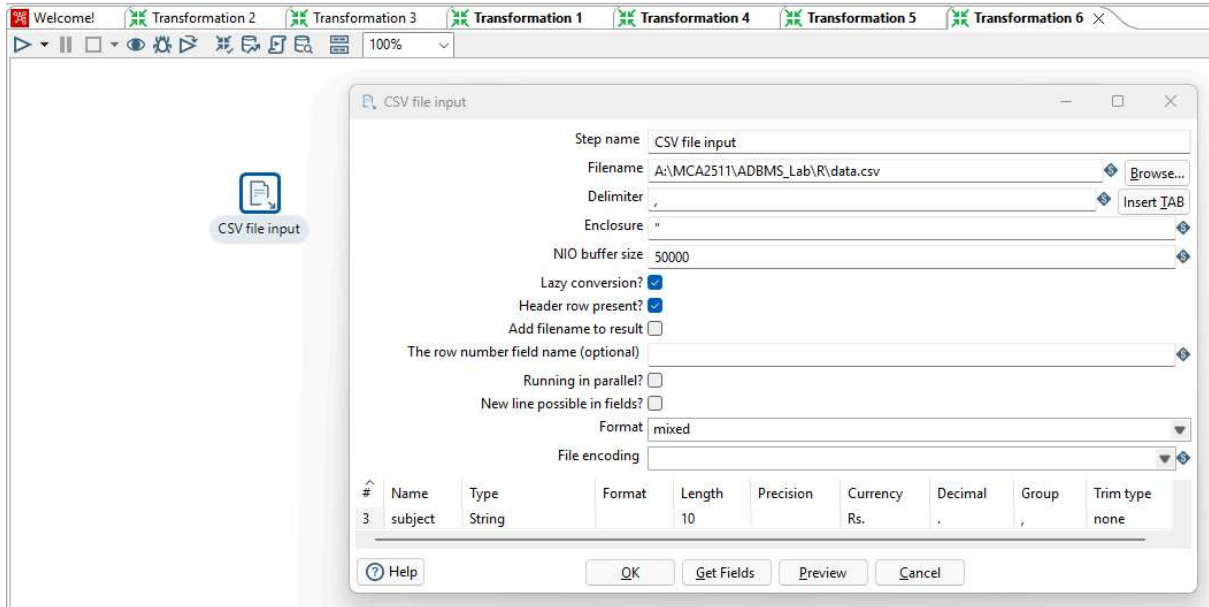
Examine preview data

Rows of step: String operations (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	**abc	xyz--	201	2006/03/15 00:00:00.000000000	40000	101
2	2	***e2	p1---	201	2016/05/15 00:00:00.000000000	30000	101
3	3	***e3	p2---	201	2005/06/28 00:00:00.000000000	20000	103
4	4	***e4	p1---	201	1972/03/05 00:00:00.000000000	15000	102
5	5	***e5	p3---	201	2016/04/25 00:00:00.000000000	35000	103
6	7	***e6	p3---	201	2016/04/25 00:00:00.000000000	15000	101
7	8	***e7	p2---	202	2018/01/12 00:00:00.000000000	28000	102
8	9	***e8	p1---	202	2015/09/03 00:00:00.000000000	32000	101
9	10	***e9	p3---	201	2019/11/20 00:00:00.000000000	36000	103
10	11	**e10	xyz--	203	2014/02/17 00:00:00.000000000	41000	104
11	12	**e11	p1---	202	2011/05/29 00:00:00.000000000	29000	101
12	13	**e12	p2---	203	2017/08/08 00:00:00.000000000	22000	102
13	14	**e13	p3---	202	2013/07/19 00:00:00.000000000	37000	104
14	15	**e14	xyz--	201	2009/12/06 00:00:00.000000000	43000	103

6. Importing csv file

Input



Output

#	name	age	subject
1	Mukund	20	python
2	Omkar	21	java
3	Aniket	23	r
4	Nimish	22	javascript
5	Atlamash	24	cpp
6	Sumit	22	c
7	Bhaves	22	sql

#	name	age	subject
1	Mukund	20	python
2	Omkar	21	java
3	Aniket	23	r
4	Nimish	22	javascript
5	Atlamash	24	cpp
6	Sumit	22	c
7	Bhaves	22	sql

7. Merge Join

Input

Data grid

Step name Employee

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Null if	Set empty string?
1	EmpId	Integer								N
2	EmpName	String								N
3	Age	Integer								N
4	Dept	Integer								N

Help OK Preview Cancel

Data grid

Step name Employee

#	EmpId	EmpName	Age	Dept
1	1001	Mukund	20	101
2	1002	Jack Ma	25	101
3	1003	Anu Shree	63	103
4	1004	Omakr	25	102

Help OK Preview

Data grid

Step name Department

#	Name	Type	Format	Length	Precision	Currency
1	DeptNo	Integer				
2	DName	String				

Help OK Preview Cancel

[illegible]

Merge join

Step name: Merge join

First Step: Sort rows

Second Step: Sort rows 2

Join Type: INNER

Keys for 1st step:

#	Key field
1	Dept

Get key fields

Keys for 2nd step:

#	Key field
1	DeptNo

Get key fields

Help OK Cancel

Output:

Examine preview data

Rows of step: Merge join (3 rows)

#	EmpId	EmpName	Age	Dept	DeptNo	DName
1	1001	Mukund	20	101	101	IT
2	1002	Jack Ma	25	101	101	IT
3	1003	Anu Shree	63	103	103	AI

8. Data Validation

Input

Data grid

Step name: Product

#	Name	Type	Format	Length	Precision	Currency	Decimal	Group	Null if	Set empty string?
1	Pid	Integer								N
2	Pname	String								N
3	Price	Integer								N
4	Delivery Status	String								N

Help OK Preview Cancel

Data grid

Step name: Product

#	Pid	Pname	Price	Delivery Status
1	101	Mouse	500	Shipped
2	102	Camera	20000	Cancelled
3	103	Lens	5000	Delivered

Help OK Preview Cancel

Data validator

Stepname: Data validator

Select a validation to edit:

Product_Validation

☐ Report all errors, not only the first

☐ Output one row, concatenate errors with separator: |

Validation description: Product_Validation

Name of field to validate: Delivery Status

Error code:

Error description:

Type

Verify data type? ☐

Data type: String

Conversion mask:

Decimal Symbol:

Grouping Symbol:

Data

Null allowed? ☒

Only null values allowed? ☐

Only numeric data expected? ☐

Max string length:

Min string length:

Maximum value:

Minimum value:

Expected start string:

Expected end string:

Not allowed start string:

Not allowed end string:

Regular expression expected to match:

Help OK New validation Remove validation Cancel

Data

Null allowed? ☒

Only null values allowed? ☐

Only numeric data expected ☐

Max string length

Min string length

Maximum value

Minimum value

Expected start string

Expected end string

Not allowed start string

Not allowed end string

Regular expression expected to match

Regular expression not allowed to

Allowed values

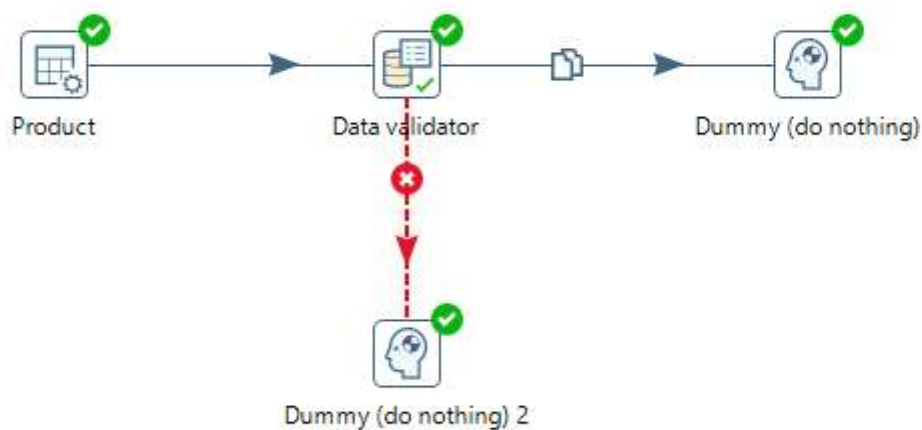
Shipped

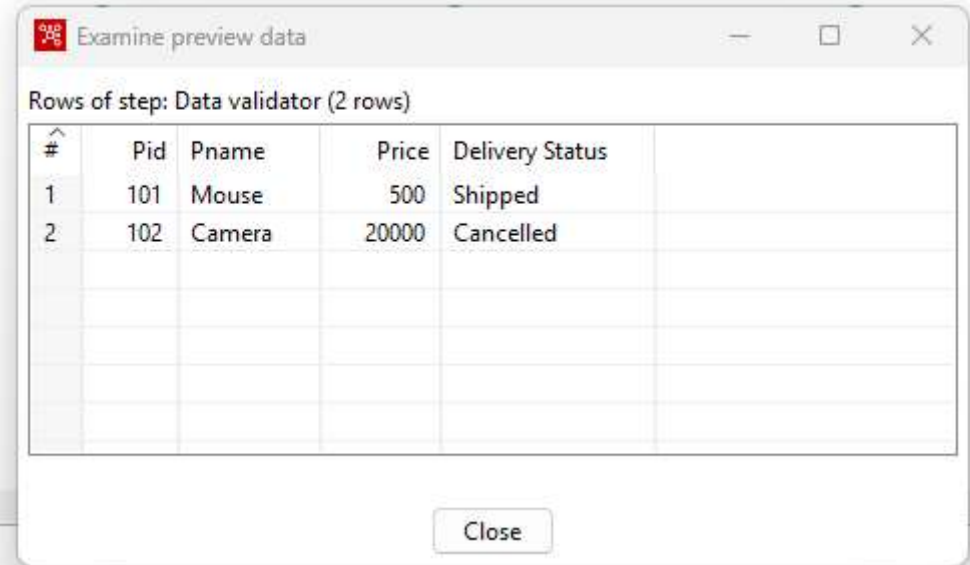
Cancelled

Read allowed values from another step? ☐

New validation Remove validation Cancel

Output





The screenshot shows a window titled "Examine preview data" with a table titled "Rows of step: Data validator (2 rows)". The table has 5 columns: "#", "Pid", "Pname", "Price", and "Delivery Status". The first two rows contain data: Row 1 has Pid 101, Pname Mouse, Price 500, and Delivery Status Shipped; Row 2 has Pid 102, Pname Camera, Price 20000, and Delivery Status Cancelled. There are four empty rows below. A "Close" button is at the bottom right.

#	Pid	Pname	Price	Delivery Status
1	101	Mouse	500	Shipped
2	102	Camera	20000	Cancelled